Dreamlike Diffusion and Dreamphotoreal

Course Name: Advanced Topics in Artificial Intelligence

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Observations and Experiments

1. It was observed that DreamBooth performs optimally when the training images have similar settings with faces aligned in comparable positions; using multiple images with divergent settings tends to confuse the model.

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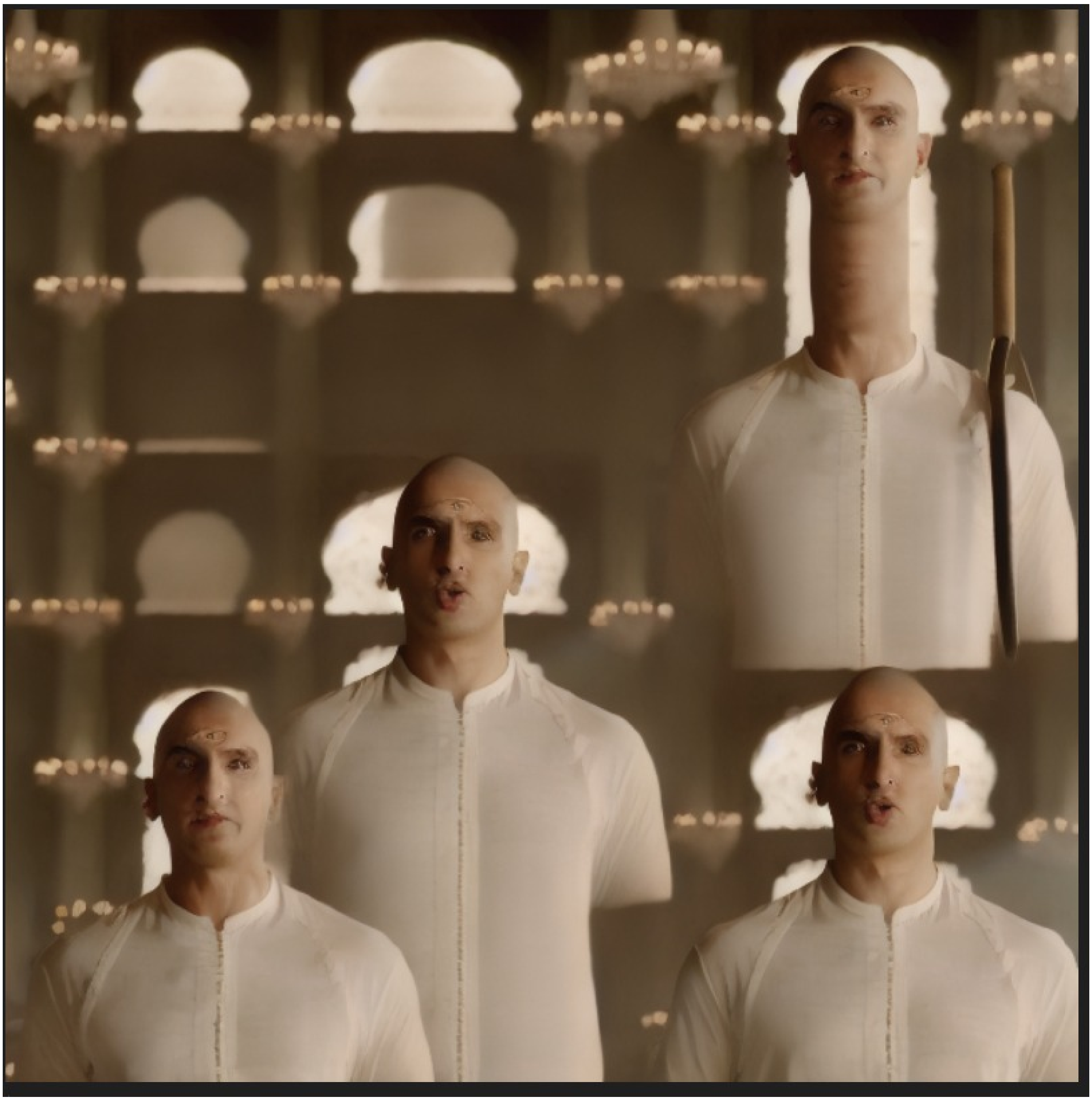
Input image used:

Output Image Demonstrating 1st Point

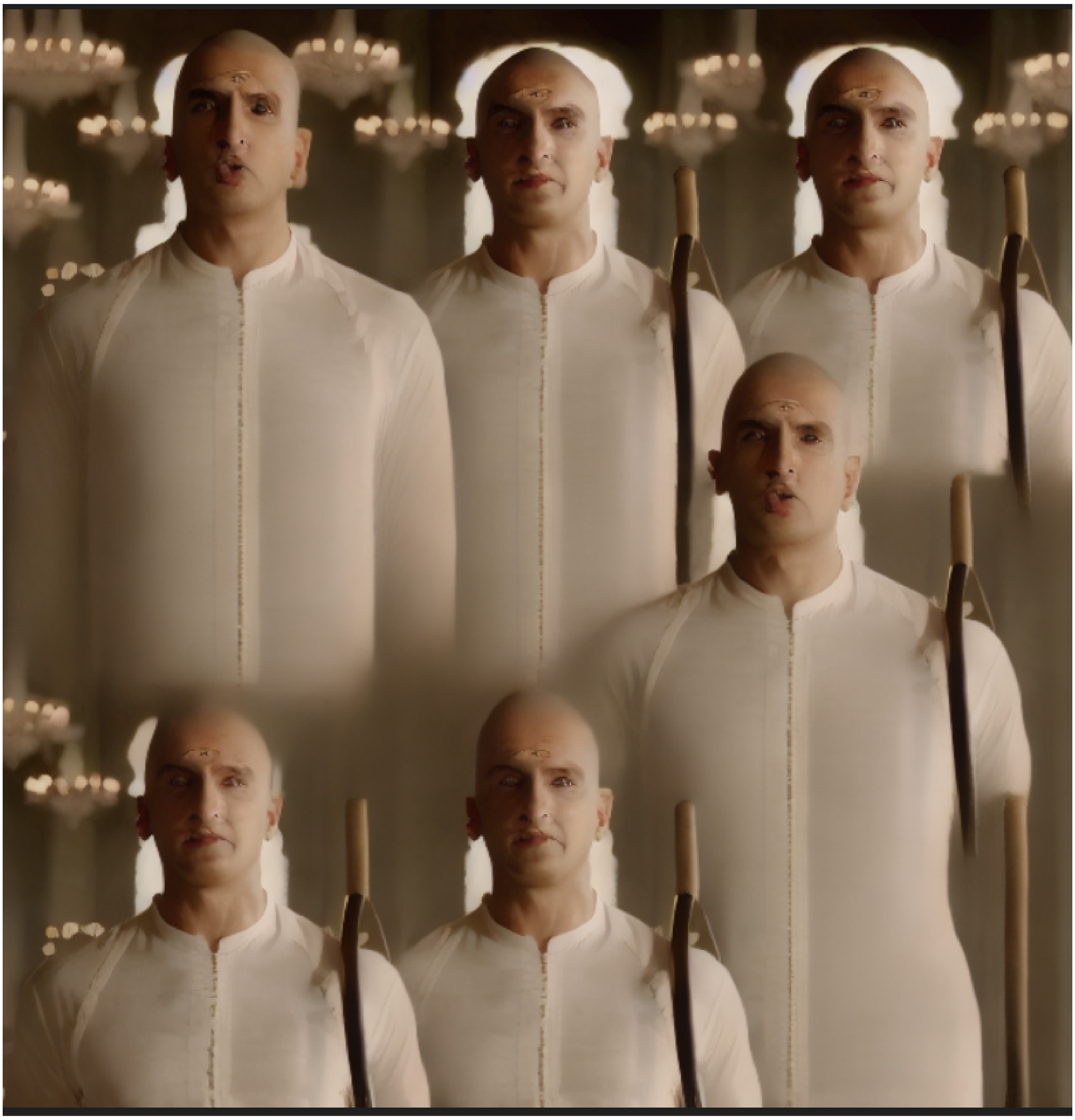


The elongated neck seems that it tried to superimpose both the second image and the third image to some extent.Because this similar generation continued for multiple generations but this elongated generated disappeared when I used similar level images. It can be seen in point 2's output.

2. Employing a token such as `<bajiface>` is advantageous maybe because the inclusion of the word "face" guides the model to focus on facial features. In contrast, very simple tokens like "r" are too generic due to their strong prior probability in the model's latent space and even complex tokens tend to confuse it in some sense like in one training prompt used was "<bajiface¿ wearing white dress" but in the output, the images generated were somewhat similar to bajirao but had white patch on his head.

Input image used:

Output using <bajiface¿



Part of the above output with superposition of the two input images using <bajiface¿

3. Both Dreamlike Diffusion and Dreamphotoreal have been pre-trained on images of Ranveer Singh, which impacts subsequent fine-tuning experiments if we use any reference to Ranveer Singh in the token. It is crucial to avoid using direct names, such as "Ranveer Singh," or tokens like `bajirao_ranveer` in training prompts because their strong priors in the model's latent space inhibit learning. Instead, unique tokens,with some attributes of the picture to be learnt, that prompt random outputs during inference should be used.

Output of inference with Ranveer Singh as input without any finetuning

4.The use of higher quality images significantly enhances the overall performance of the model during Dreambooth.

5. Even after fine-tuning via DreamBooth, when prompted to generate one image of Ranveer Singh and two images of different subjects, the model consistently produces three images that all resemble Ranveer Singh.

6. Experiments with alternative VAEs reveal that for using different VAE we need to train the VAE properly with a larger dataset first since on lesser number of images it gave nearly a noisy image.

7. Training the text encoder along with DreamBooth negatively affected the results, suggesting that decoupling this training component may be beneficial while using dreambooth.

8. A practical metric for evaluating the fine-tuned model's performance seems to be only using the unique token as a prompt—such as by writing "photo of `<unique-token>`"—and then assess whether the output reflects the intended style after fine-tuning.

9. Modifying and training the VAE during Dreambooth fine-tuning did not yield acceptable results when only a few training steps were used, although a larger dataset and longer training duration might improve performance.

10.Using only one image where Ranveer Singh is there with a glorified background can also disrupt the training process by confusing the model about whether it should focus on the subject's face or background because on multiple trials the token`<ranbaji>` was just generating background for one image of Ranveer Singh. Maybe this can be overcome, using tokens such as `<bajiface>` and using more focused image of Ranveer Singh (Two experiments seems to be confirming it while using my own face or by using Ranveer's face but still need to check the consistency).

11. When my own face was used for training, the model was able to replicate my facial features accurately, and it even managed to include additional elements like a crown on the head to some extent.But the consistency was a issue ,maybe because I had used only smaller training steps. Need to check for more training steps once.

Input of My face

Output for the token only

Output for the prompt : `<bajiface> with a crown on head`

Note: With my image,since my image was totally focused on my face, the output was better even using one image.Still a confusing fact that for my image it's able to replicate it once but in case of Ranveer Singh it's generating multiple Ranveer in one image only.

0.1 Prompts used for inferring the model:

bajirao1: a cinematic photo of `<baji-face>` from *Bajirao Mastani*
bajirao2: a dramatic photo of `<baji-face>` wearing Maratha armor
bajirao3: a movie still of `<baji-face>` in a battlefield scene
bajirao4: a historical portrait of `<baji-face>` with royal attire
bajirao5: a close-up of `<baji-face>` with a fierce expression
bajirao6: a dynamic shot of `<baji-face>` as Bajirao
bajirao7: a royal photo of `<baji-face>` with a sword

12. The model tends to generate better-aligned outputs when it is trained totally on a carefully curated larger dataset of Ranveer Singh, ensuring prompt coherency with the training images (observed just by inferencing dreamlike models on Ranveer Singh prompts and no fine-tuning).

Output of inference with original model

bajirao1

bajirao2

bajirao3

bajirao4

bajirao5

bajirao6

bajirao7

13. Using textual inversion with Dreamlike Diffusion and Dreamphotoreal with even lesser number of steps gives somewhat similar features as Ranveer but not exactly Ranveer. Advantage is that it is able to give only one face rather than multiple face but images are not exactly alike.

Output of inference with fine-tuned model on Textual Inversion

bajirao1

bajirao2

bajirao3

bajirao4

bajirao5

bajirao6


bajirao7